

ECON 547—ADVANCED GAME THEORY: MECHANISM DESIGN

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Department of Economics
Nazarbayev University

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Lecture time/Location: TBA

Office hours: By appointment.

Course Description This is an advanced course in Game theory, with a focus on the theory of Mechanism Design, which is a subfield of economics that aims to construct economic mechanisms (rules, environments, institutions) with desirable properties. While traditional microeconomic analysis seeks to understand how agents behave within established frameworks, mechanism design aims to design ‘suitable’ rules that lead to ‘desirable’ outcomes.

The course is divided into two halves. In the first half, delivered in a lecture/seminar format, we will cover the standard results in the theory of mechanism design, and the foundations of Incomplete Information and Robustness in Games. Additionally, we will delve into topics such as Information Design, robust mechanisms, obviously-strategy-proof mechanisms, and mechanism design under bounded rationality.

The second half of the course adopts a tutorial/discussion group style. During this phase, we will explore state-of-the-art research, with the specific topics to be discussed determined by student and instructor’s interests. Possible areas of focus include dynamic mechanism design, endogenous information provision, hidden action, limited commitment, and revenue management.

Course Requirements: ECON501 Microeconomics 1 (B and above), ECON512 Mathematical Economics (B and above), ECON521 Microeconomics 2 (B and above)

Course Aims:

1. Understand the standard topics in Mechanism Design
2. Understand the formal tools required to model incomplete information and common knowledge of rationality in games, as well as the ideas of robustness of a game-form to changes in the informational environment.
3. Analyze applications of the previous two aims to Information Design, robust mechanisms, and the design of mechanisms for boundedly rational agents.
4. Get in-depth knowledge of the state-of-the-art literature in Mechanism Design.
5. Propose a research idea, conduct the literature review, analyze its merits, and propose the mathematical framework required to conduct the analysis.
6. Write and present a draft of a research paper, including the components outlined in aim (5) as well as a description of the mathematical model, and at the minimum, a fully solved example of the model.

Course Learning Outcomes: By the end of the course the student will be expected to be able to

1. Demonstrate in-depth knowledge of Mechanism Design as well as the modeling of incomplete information and robustness in strategic situations.

2. Demonstrate in-depth knowledge of the state-of-the-art literature in Mechanism Design.
3. Be able to synthesize the recent literature on a topic of interest, propose a research project, develop the mathematical framework to carry out the analysis, and develop and analyze a solution to the proposed model.
4. Communicate effectively, both in oral and written form, the results of the research carried out.

Learning and Teaching Methods: The first half of the course will be conducted in a lecture/seminar format. The students are expected to read and study all assigned material before each session and come prepared to discuss it. The instructor will briefly cover the key concepts but most of the learning will be through a deep discussion of the material in which the student should be the main participant. The instructor will assign from time-to-time problems/exercise to challenge the understanding of the students and to incentivize discussion for the next sessions.

The second half of the course will be conducted in a tutorial/discussion group format. Students are expected to read recent papers, either proposed by them or the instructors, and be ready to discuss them in the sessions.

Course Textbooks: There is no textbook required for this course. All materials will be uploaded to Moodle.

Course Website: All required material will be posted to Moodle

Assessments

Research Proposal	25%
Presentation	25%
Research paper draft	50%

Research Proposal The student should propose a research project. It can be theoretical or empirical, but related to the class material. The proposal is at most 5 pages long. It should state the question, its merits, and a literature review. An example highlighting the desired result is ideal, but not necessary.

Presentation AT the end of the class, there will be short presentations of the advanced made by the students in their research projects.

Research Paper A write-up documenting what was already described in the research proposal, i.e., the question, merits and literature review, as well as a description of the model and mathematical framework used to carry out the analysis. The paper should contain at the very minimum a fully solved example of the model.

I will compute your final numeric grade by computing the weighted average of the aforementioned components according to the weights given in the table above.

Grading: Your course grade is determined by the weighted average of the assessments above according to the following scale:

A	95–100	A–	90–94.99
B+	85–89.99	B	80–84.99
B–	75–79.99	C+	70–74.99
C	65–69.99	C–	60–64.99
D+	55–59.99	D	50–54.99
F	0–44.99		

Course Expectations:

1. Students are expected to perform the assigned reading and activities before the session so that they are prepared to engage vigorously in the discussion of the topics.
2. In the beginning the first sessions will have a more standard lecture format but participation from the students will be crucial.
3. As the course progresses, the class will shift progressively to a tutorial/discussion format, in which the instructor will lead the discussion, but the students are expected to engage in a deep analysis and discussion of the assigned readings/topics.
4. This is a course primarily for students interested in doing research in economic theory as part of their MA thesis writing efforts.
5. The course is advanced and intended for students with serious interests in economic theory. Students who are not interested in theory research and do not have the appropriate prerequisite mathematical and economic background are strongly recommended to not enroll in this course.

Academic Misconduct: Students are expected to act with integrity. Cheating and plagiarism will not be tolerated. Any student deemed academically dishonest will be reported to the SSH disciplinary committee. Any cases of academic misconduct will be handled in accordance with the procedures of the Student Code of Conduct. The penalties may include anything from failing grade on a particular assignment to a loss of stipend, and even, dismissal from the University. You can read more about academic misconduct in the Student Code of Conduct. https://aao.nu.edu.kz/academic_integrity

Tentative Course Outline:

1. Foundations of Mechanism Design and Implementation
 - (a) Dominant-Strategy mechanism Design
 - i. Definitions
 - ii. Revelation Principle
 - iii. The Gibbard-Satterthwaite Theorem
 - iv. Groves' Mechanisms
 - v. Problems of Balancedness and Individual Rationality in Groves' Mechanisms
 - vi. Inefficient Decisions
 - (b) Nash Implementation
 - i. Maskin Monotonicity
 - ii. Main Results
 - iii. Limitations
 - (c) Bayesian Mechanism Design
 - i. Revelation Principle
 - ii. A balanced Mechanism with Independent-Types: the Arrow d'Aspremont and Gerard-Varet mechanism.
 - iii. The Myerson and Satterthwaite Theorem.
 - iv. Correlated Types and Surplus Extraction
2. Foundations of Incomplete Information
 - (a) Common Knowledge and Rationality
 - i. Agree to Disagree
 - ii. Rubinstein's email game

- (b) Incomplete Information and Robustness
 - i. Harsanyi Type Spaces and the Universal Belief Space.
 - ii. Bayes-Nash Equilibrium and Concepts of Rationalizability
 - iii. Bayes Correlated Equilibrium

3. Information Design

- (a) Bayesian Persuasion
- (b) Psychological Preferences and Bayesian Persuasion (tentative)
- (c) Information Design with Many Agents

4. Advanced Mechanism Design

- (a) Robustness and Robust Mechanisms.
- (b) Bounded Rationality in Mechanism Design.
- (c) Obviously-Strategy-Proof Mechanisms.

5. Tutorial/Class Discussions on state-of-the-art literature. All topics are tentative. A subset of the topics will be selected subject to student interest. Other topics can be covered at the request of students.

- (a) Communication Games.
- (b) Revenue Management.
- (c) Incentive compatibility in multidimensional environments.
- (d) Applications to Auctions, matching, hidden-action and self-control.
- (e) Sequential Voting.
- (f) Dynamic Information Design.
- (g) Experts and Delegation.

Session	Topics and Assignments	Course Aims	CLOs
Week 1	Dominant-Strategy Mechanism Design	1	1
Week 2	Nash Implementation	1	1
Week 3	Bayesian Mechanism Design	1	1
Week 4	Common Knowledge and Rationality	2	1
Week 5	Incomplete Information and Robustness	2	1
Week 6	Information Design	3	1,2
Week 7	Advanced Mechanism Design: Robustness and Bounded Rationality	3	1,2
Week 8	Fall Break/Spring Break		
Week 9–13	Discussion sessions/tutorials on state-of-the-art literature	4	3,4
Week 11	Delivery of Research Proposal	5,6	3,4
Week 14–15	Student research papers' discussions and presentations	5,6	3,4

Disclaimer Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on Moodle.